# AI Chips, Energy & Nuclear Index Dashboard: A Power BI–Driven Analytical Framework


## Abstract

The rapid growth of artificial intelligence (AI) technologies has led to a surge in demand for high-performance computing infrastructure, particularly AI chips. This demand has significant implications for energy consumption and has renewed interest in nuclear energy as a sustainable power source. This project presents an interactive Power BI dashboard titled *"AI Chips, Energy & Nuclear Index Dashboard"*, designed to analyze and visualize relationships between AI chip markets, energy consumption trends, and nuclear energy indices. The dashboard integrates multi-source datasets to provide actionable insights for investors, policymakers, and industry stakeholders. The study demonstrates how data visualization tools can bridge complex interdependencies across technology and energy sectors.


## 1. Introduction

The expansion of AI applications ranging from machine learning to generative AI—has significantly increased the need for specialized hardware such as GPUs, TPUs, and custom AI chips. These technologies require substantial computational power, which in turn drives high energy consumption.

Simultaneously, concerns about sustainability and carbon emissions have pushed industries to explore cleaner energy alternatives. Nuclear energy, with its low carbon footprint and high efficiency, is emerging as a viable solution to meet the energy demands of AI infrastructure.

This project aims to explore the intersection of:

- AI chip industry growth
- Global energy consumption trends
- Nuclear energy development

Using Power BI, the project transforms raw datasets into a comprehensive analytical dashboard.

## 2. Methodology

To achieve the project objectives, the following steps were undertaken:

1. **Data Integration**
   Multiple datasets from different domains were combined into a unified model.
2. **Data Modeling in Power BI**
   - Created relationships between tables
   - Developed calculated columns and measures using DAX

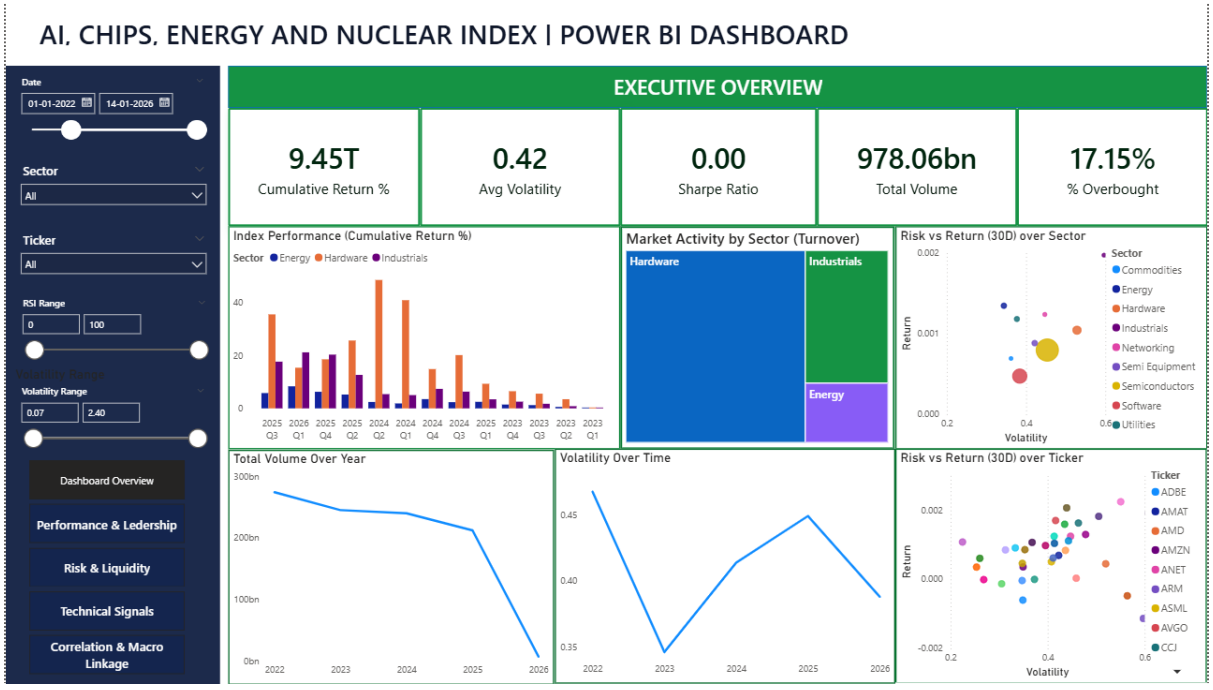
- Designed star schema for efficient querying

3. **Visualization Design**
    - Built dashboards with charts such as line graphs, bar charts, and heat maps
    - Implemented KPIs for quick insights
    - Used slicers for dynamic filtering
4. **Insight Generation**
    - Identified trends and correlations
    - Highlighted key growth areas and risks
    - Enabled comparative analysis across sectors

## 3. Business Findings & Analytical Explanation

### A. Market Leadership & Growth

- Focus on Relative performance + trend leadership. Bar charts (time-series trends) comparing AI chip companies, Energy sector, Nuclear index. KPI cards showing total returns and growth percentages.
- AI chip companies consistently display strong upward trends, especially during periods of increased AI adoption.
- Energy and nuclear sectors show more stable but slower growth trajectories.
- Relative performance visuals clearly highlight that: AI chips often outperform traditional energy sectors in bullish periods and nuclear begins to gain traction during sustainability-focused phases.

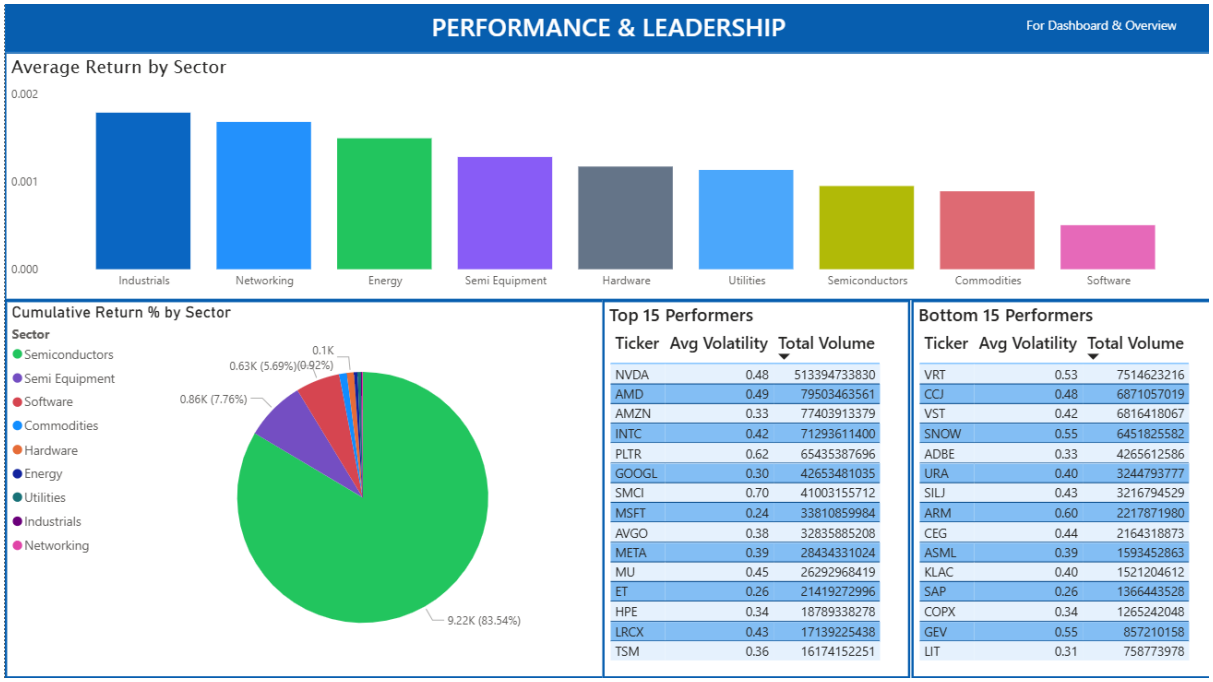


**B.** Risk & Volatility

- Focus on risk-return trade-off. This is analysed through Scatter plots (Risk vs Return) and Volatility metrics.
- The dashboard confirms a classic risk-return tradeoff, the higher returns in AI chips come with higher risk exposure.
- Volatility spikes in AI companies coincide with sharp returns (both gains and corrections).
- Energy/nuclear act as defensive sectors, stabilizing portfolios.

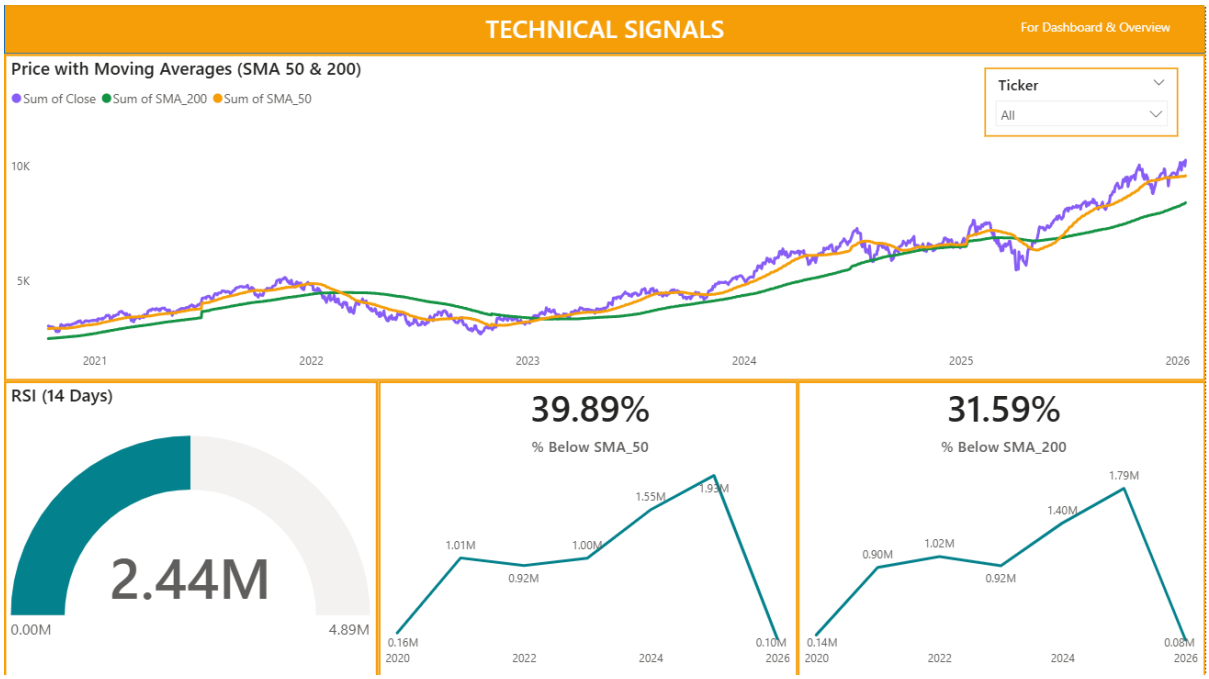
**C. Liquidity & Market Activity**

- Focus on market attention + liquidity. Analyzed using volume and turnover charts and sector-wise liquidity comparison visuals.
- It can interpretation that Capital is clearly flowing toward AI-related assets. High liquidity in AI industry indicates Strong investor confidence and Speculative and institutional interest. Energy and nuclear sectors are show consistent but lower liquidity.



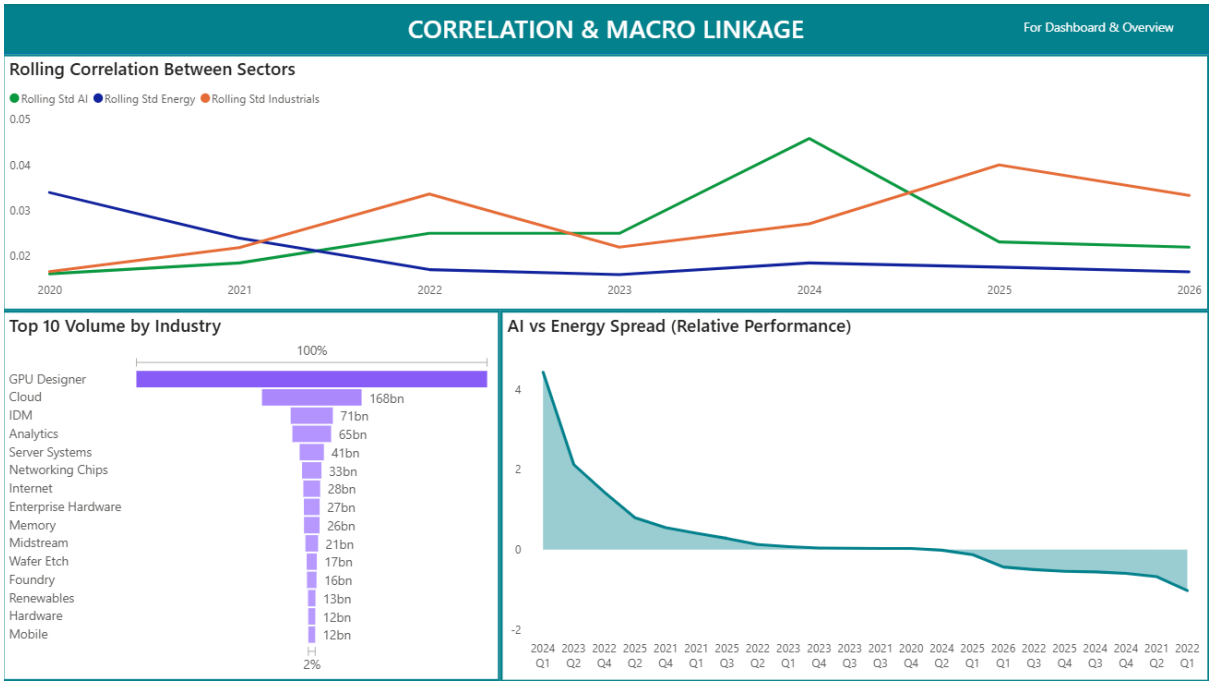

## D. Technical Momentum

- Focus on Timing signals (RSI & SMA). Visualized through RSI (Relative Strength Index) indicators and Moving averages (SMA crossover charts).
- AI chip stocks frequently enter overbought zones (RSI > 70) during rallies.
- Energy stocks stay within neutral RSI ranges.
- SMA crossover signals that AI sector shows frequent bullish crossovers and energy/nuclear show gradual, stable trends.



**E. Sector Interdependency**

- Focus on Correlation & macro linkage. This is analyzed using Dual-axis trend comparisons and Overlay charts (AI vs Energy/Nuclear).
- The dashboard highlights a clear macro relationship: Growth in AI → increased demand for power → impact on energy markets.
- Nuclear energy emerges as a strategic long-term solution for AI-driven energy needs.




By integrating multiple visual layers, we can conclude that:

- Identifies AI chips as growth leaders
- Highlights risk concentration in AI vs stability in energy
- Tracks capital flow toward high-growth sectors
- Provides technical indicators for timing decisions
- Reveals interdependency between compute demand and energy supply

## 4. How This Helps Industry

- **Investors** → Better portfolio allocation (growth vs stability)
- **Energy companies** → Anticipate demand from AI infrastructure
- **AI firms** → Understand dependency on energy supply chains
- **Policy makers** → Plan sustainable energy strategies for AI expansion

## 5. Future Scope

- Integration of real-time data feeds
- Inclusion of renewable energy sources (solar, wind)
- Advanced predictive analytics using machine learning
- Expansion to global policy impact modeling

## Conclusion

The *"AI Chips, Energy & Nuclear Index Dashboard"* demonstrates how data visualization tools like Power BI can effectively analyze complex relationships between technology and energy sectors. By integrating AI chip trends with energy consumption and nuclear energy data, the project provides valuable insights that can guide industry strategies, investment decisions, and policy development.

As AI continues to evolve, the need for sustainable and scalable energy solutions will become increasingly critical. This dashboard serves as a foundational tool for understanding and navigating this interconnected landscape.